LIGHTSHIP WEIGHT ESTIMATION BOOKLET

Vessel Name:	M.V. CROWN MARINERS - 67	Operating Route:	Inland & bay crossing
Vessel Type:	Steel Body Cargo Vessel	Plying Time:	Day & Night
Hull Construction:	Single Hull	Statutory Factor kt / kz:	1.40 / 1.40
Length Over All (LOA):	76.21 m	Breadth Moulded (B):	13.41 m
Length BP (LBP):	69.85 m	Depth Moulded (D):	5.18 m
Design Draft (d):	4.00 m	Block Coeff (Cb):	0.721
Transverse Frame Spacing:	550 mm	Structural Steel Density:	7,850 kg/m³

NAVAL ARCHITECTURE DESIGNER	INSPECTION & APPROVAL AUTHORITY
ROUNAK SAHA NILOY Naval Architect and Marine Engineer Designer ShipDyn Ltd.	Amitavo Chowdhury Wye Authorized Person ShipDyn Ltd.

1. STEEL PLATING WEIGHT ESTIMATION

Scantlings Plating Volume and Weight Evaluation | Mild Steel Density: 7,850 kg/m³

S/N	ELEMENT (Clean Plate Name)	L (mm)	B (mm)	t (mm)	Vol (m³)	Density	Qty	Weight (kg)	Weight (t)
1	Keel Plate	76,210	600	12.0	0.5487	7850	1	4,307	**4.307**
2	Bottom Shell Plating	76,210	4,605	11.0	3.8604	7850	2	60,609	**60.609**
3	Bilge Plating	76,210	3,000	14.0	3.2008	7850	2	50,253	**50.253**
4	Side Shell Plating	76,210	3,680	12.0	3.3654	7850	2	52,837	**52.837**
5	Sheer Strake Plating	76,210	900	12.0	0.8231	7850	2	12,922	**12.922**
6	Bulwark Plating	50,600	1,000	6.0	0.3036	7850	2	4,767	**4.767**
7	Hatchway Deck Plating	47,740	2,100	13.0	1.3033	7850	2	20,462	**20.462**
8	Main Deck Plating	17,350	11,928	9.0	1.8626	7850	1	14,621	**14.621**
9	Main Deck Wall	15,753	2,300	12.0	0.4348	7850	1	3,413	**3.413**
10	Poop Deck Wall	8,803	2,100	12.0	0.2218	7850	1	1,741	**1.741**
11	Poop Deck Plating	15,499	9,781	9.0	1.3644	7850	1	10,710	**10.710**
12	Bridge Deck Wall	3,100	2,000	12.0	0.0744	7850	1	584	**0.584**
13	Bridge Deck Plating	7,840	8,803	9.0	0.6211	7850	1	4,876	**4.876**
14	Transverse Watertight Bulkheads	13,410	5,180	10.0	0.6946	7850	5	27,265	**27.265**
15	Cargo Hatch Covers Plating	14,630	8,993	13.0	1.7104	7850	3	40,279	**40.279**
16	Forecastle Deck Plating	10,111	8,516	9.0	0.7749	7850	1	6,083	**6.083**
17	Double Bottom Tank Top Plate	76,210	13,410	0.0	0.0000	7850	0	0	**0.000**
18	Stem, Stern & Rudder Assembly	8,000	3,500	16.0	0.4480	7850	2	7,034	**7.034**
19	Superstructure Partition Bulkheads	13,410	2,300	12.0	0.3701	7850	2	5,811	**5.811**
	TOTAL STEEL PLATING WEIGHT:							**328,574 kg**	**328.574 t**

2. STEEL STIFFENERS & STRUCTURAL ANGLES WEIGHT ESTIMATION

Stiffener Scantlings Geometry & Unsupported Span Evaluation (Length linked to attached plate)

S/N	ELEMENT (Structural Profile)	L (mm)	B (mm)	t (mm)	Area (mm²)	Length	Vol (m³)	Density	Qty	Weight (kg)	Weight (t)
1	Center Keelson	500	100	12.0	7,200	76,210	0.5487	7850	1	4,307	**4.307**
2	Side Keelson	350	100	10.0	4,500	76,210	0.3429	7850	8	21,537	**21.537**
3	Side Stringer	250	100	10.0	3,500	76,210	0.2667	7850	6	12,563	**12.563**
4	Deck Girder	450	150	11.0	6,600	76,210	0.5030	7850	7	27,639	**27.639**
5	Floor	500	180	12.0	8,160	6,705	0.0547	7850	126	54,116	**54.116**
6	Web Frame	350	120	11.0	5,170	4,680	0.0242	7850	27	5,128	**5.128**
7	Main Frame	200	100	12.0	3,600	4,680	0.0168	7850	100	13,226	**13.226**
8	Deck Beam	125	75	9.0	1,800	6,705	0.0121	7850	52	4,927	**4.927**
9	Deck Web	300	150	10.0	4,500	6,705	0.0302	7850	27	6,395	**6.395**
10	Deck Beam (Main Deck)	75	75	6.0	900	6,705	0.0060	7850	30	1,421	**1.421**
11	Deck Beam (Poop Deck)	65	65	6.0	780	3,813	0.0030	7850	18	420	**0.420**
12	Deck Beam (Bridge Deck)	50	50	5.0	500	2,897	0.0014	7850	7	80	**0.080**
13	BHD Angle (Vertical)	120	120	10.0	2,400	4,680	0.0112	7850	45	3,968	**3.968**
14	BHD Angle (Horizontal)	120	120	10.0	2,400	13,410	0.0322	7850	25	6,316	**6.316**
15	Hatch Girder	65	65	6.0	780	4,605	0.0036	7850	15	423	**0.423**
16	Superstructure Girder (Main Deck)	75	75	6.0	900	2,300	0.0021	7850	31	504	**0.504**
17	Superstructure Girder (Poop Deck)	65	65	6.0	780	2,100	0.0016	7850	17	219	**0.219**
18	Superstructure Girder (Bridge Deck)	50	50	5.0	500	2,000	0.0010	7850	6	47	**0.047**
19	Superstructure Long. Girder (Main Deck)	200	90	7.0	2,030	15,753	0.0320	7850	1	251	**0.251**
20	Superstructure Long. Girder (Poop Deck)	200	90	7.0	2,030	8,803	0.0179	7850	1	140	**0.140**
21	Superstructure Long. Girder (Bridge Deck)	200	90	7.0	2,030	3,100	0.0063	7850	1	49	**0.049**
	TOTAL STEEL STIFFENERS WEIGHT:									**163,676 kg**	**163.676 t**

3. STATUTORY LIGHTSHIP SYNTHESIS & TOTAL WEIGHT ESTIMATION

Weight Item / Statutory Allowance Category	Basis Formula	Weight (Tonne)
Total Net Structural Steel (Plates + Angles)	W_plates + W_angles	**492.251 t**
Hull Allowance / Rolling & Welding Margin (20%)	Net Steel × 20%	98.450 t
Outfit & Deck Equipment (15%)	Net Steel × 15%	73.838 t
Piping, Valves & Fittings (10%)	Net Steel × 10%	49.225 t
Electrical & Instrumentation (3% formula / 5% rule base)	Net Steel × 3%	14.768 t
Accommodation Fittings & Furniture (1%)	Net Steel × 1%	4.923 t
Miscellaneous Statutory Items (7%)	Net Steel × 7%	34.458 t
TOTAL LIGHTSHIP WEIGHT (LIGHTWEIGHT DISPLACEMENT) :	**Net Steel × 1.56 (56% Outfitting)**	**767.911 Tonne**

